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AI Gateway in the Cloud Native AI Landscape



Jirayut Nimsaeng (Dear)

CEO & Founder, Opsta (Thailand) Co.,Ltd.

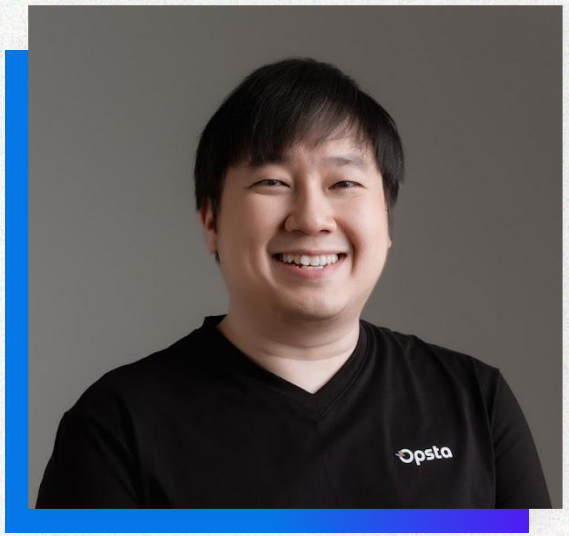
Cloud Native Bangkok: Cloud Native AI

AWS Office, 16th Floor, Singha Complex

28 August 2026

#whoami

Jirayut Nimsaeng (Dear)

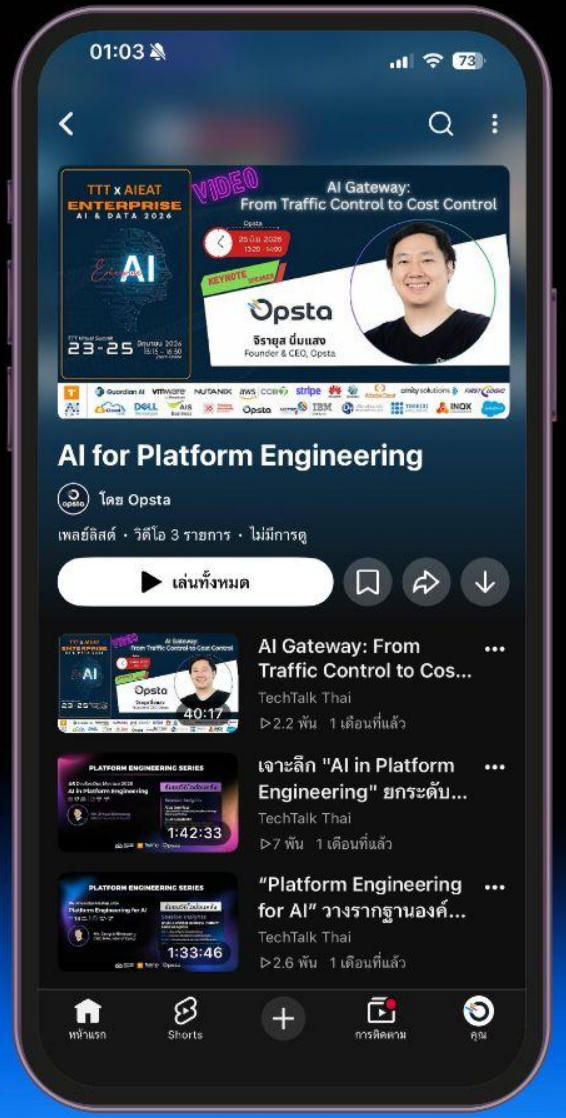


Jirayut has been involved in DevSecOps, Container,
Cloud Technology and Open Source for over 10 years.
He has experienced and succeeded in transforming several
companies to deliver greater values and be more agile.

- **Founder and CEO of Opsta (Thailand) Co.,Ltd.**
- **AI for Platform Engineering Practitioner**
- **Cloud/DevSecOps Transformation Consultant and Solution Architecture**
- **First Certified Kubernetes Administrator (CKA) and Certified Kubernetes Security Specialist (CKS) in Thailand**
- **First Thai Google Cloud Developer Expert (GDE) in Thailand**
- **Google Cloud Certified - Professional Cloud Architect and Associate Cloud Engineer**

WATCH ON-DEMAND

https://bit.ly/4qtiB8z



DevSecOps Community Thailand

TechTalk THAI

Opsta

AGENDA

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01

The Theory

- What makes AI traffic different
- Gartner's definition and required features
- Model, MCP and agent gateways
- How to evaluate one

02

The Landscape

- The AI Native Infra Gateway category
- Six projects, strengths and risks
- LiteLLM, Portkey, 9Router and the incumbents
- One feature matrix for all of them

03

The Choice

- A mental model in four layers
- A decision tree for your situation
- A reference architecture
- What to take home

PART 01

The Theory



What you need for AI Gateway

Cost is per token

Usage priced by token. One retry loop or runaway agent becomes a five figure bill before anyone opens a dashboard.

The key is the perimeter

A leaked provider key hands an attacker both your private data and an unbounded spend limit.

The backend is plural

Many providers, many models, new versions monthly. Applications should not carry that churn in their code.

The risk is in the payload

Prompt injection, PII and unsafe output live in the body. A header level proxy cannot see any of it.

An AI gateway is an API gateway that understands tokens, models and prompts, the three things a normal proxy is blind to.

Source Gartner, Market Guide for AI Gateways, 13 October 2025, ID G00839683

What Gartner means by AI gateway

> “An intermediary between applications and various AI services or models, providing a central point to enable security, governance and observability of AI workloads.”

70%

of teams building multimodel apps will use one by 2028, from 25% in 2025

< 25%

of the target audience has deployed one. An early market

$50–100M

market size in 2025, with blurred borders against API gateways

34% vs 8%

top AI performers use a gateway, against 8% of low performers

Read the last number twice. The gateway separates teams who ship AI from teams who talk about it.

Source Gartner, Market Guide for AI Gateways, 13 October 2025, ID G00839683

The five mandatory features

Access control

Who may call which model. OAuth and OIDC, RBAC, audit log.

Cost management

Token metering and reporting. Rate limits and quotas.

Multimodel routing

One interface to many models, with load balancing and failover.

Monitoring

Health checks, alerts, end to end tracing, usage dashboards.

Security

Provider keys held centrally. Data scanning and injection defense.

Nothing here is about model quality. The floor is governance, not intelligence.

Source Gartner, Market Guide for AI Gateways, 13 October 2025, ID G00839683

The eight common features

Model selection

Routing that picks the model from context.

Billing integration

Feeds usage into chargeback.

Caching

Response and semantic caching.

Data enrichment

Adds context, transforms payloads.

Guardrails

Blocks prompts, redacts sensitive data.

Retry and fallback

Survives a provider outage.

Agent protocols

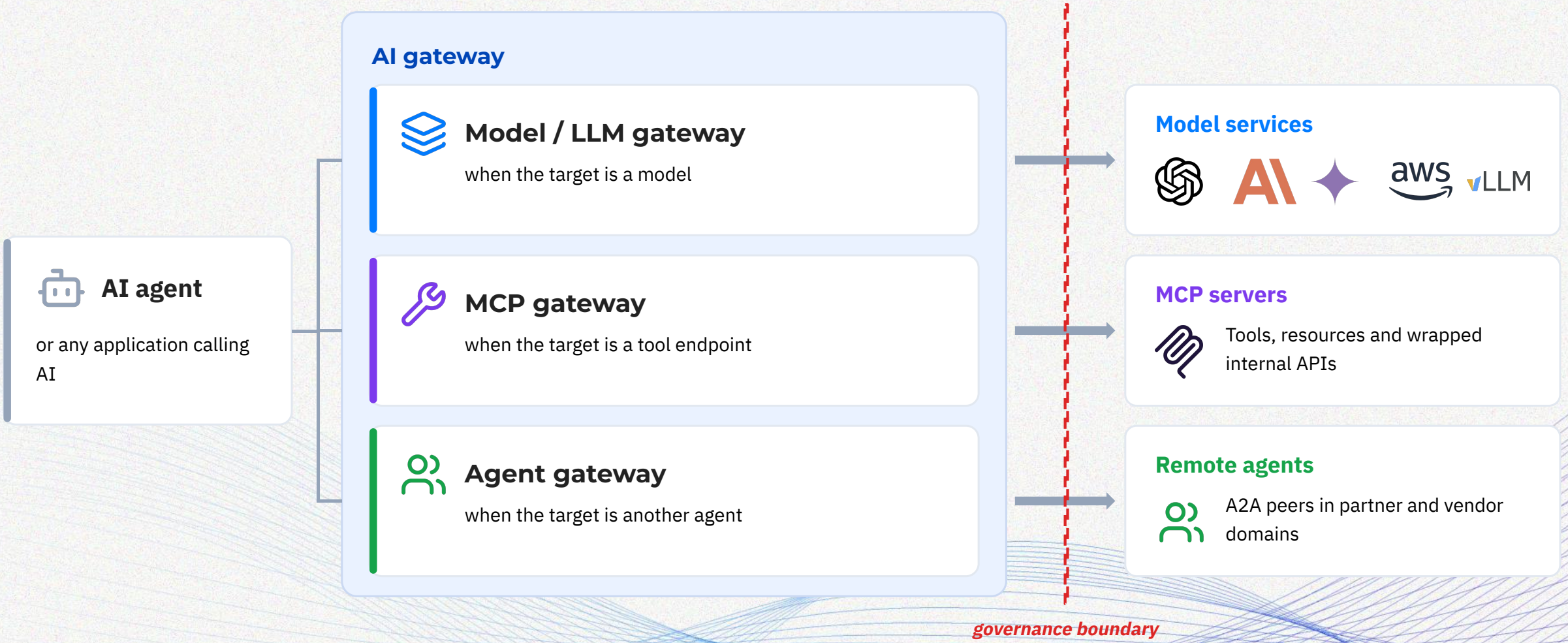
One control point over many MCP servers.

Non LLM services

Image, speech and other AI APIs.

Source Gartner, Market Guide for AI Gateways, 13 October 2025, ID G00839683

Gartner AI Gateway Architecture



Anatomy of a model / LLM gateway

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Integration modes

A unified OpenAI compatible API that translates both ways, or passthrough that forwards the provider payload untouched. Most deployments need both.

Authn and authz

Validate the caller against your IdP, apply role and model level policy, then inject the provider credential at runtime.

Security and guardrails

Prompt injection defense, PII detection and redaction, content moderation, on the request and on the response.

Cost management

Meter input and output tokens, attribute spend by app and environment, enforce quotas and hard budgets.

Routing

Model alias, metadata and traffic split routing, load balancing, fallback, and semantic routing by topic or complexity.

Prompt level features

Prompt caching, prompt decoration that injects governance context, and prompt rewriting or compression.

Source Gartner, Assessing AI Gateways to Support Agent Architecture, 15 July 2026, doc 856252

Anatomy of an MCP gateway and an agent gateway

MCP gateway · agent to tool

- Three shapes: proxy one MCP server, bridge a REST API into MCP tools, or aggregate backends into one virtual endpoint
- Two sided auth: validate the inbound token, inject the outbound credential by policy
- Tool policy at both moments, filtering tools/list and checking tools/call
- Tool poisoning and indirect injection defense, pinned definitions, DLP
- Semantic tool search, so a client gets five relevant tools not two hundred

Agent gateway · agent to agent

- Mediates A2A transport between a client agent and a remote agent
- Agent discovery through agent card rewriting and registry integration
- Identity per agent, so a peer is a principal not a shared service account
- Policy at the boundary: rate limits, token budgets, guardrails on task execution
- Observability with A2A specific trace attributes

Source Gartner, Assessing AI Gateways to Support Agent Architecture, 15 July 2026, doc 856252

AI Gateway Maturity

Obstacles to adoption

- Immaturity. Scope and maturity vary wildly between products
- Latency. AI is delay sensitive and a gateway is another hop
- Caching. Semantic caching is not HTTP caching, and errors cost accuracy
- Alternatives. Model routers and the LLM providers already do some of this

Where the market is heading

- API management platforms and AI gateways are converging
- Model routers, MCP and later A2A gateways consolidate into single products
- No policy standardization soon, so portability stays limited
- Deployment spreads across cloud, on premises, edge and hybrid

Source Gartner, Market Guide for AI Gateways, 13 October 2025, ID G00839683

Eight questions to ask any AI gateway

Gartner's evaluation criteria, reordered so the ones people forget come first

Fits what you run

Sits beside your API management, or demands its own island?

Roadmap

Commitment to evolving standards, not just today's features.

Wider AI security

Works with runtime security, testing and posture management.

Security

It becomes the access layer for sensitive data.

Regulation

Data protection, identity and risk governance duties.

Scale and latency

High concurrency without latency you cannot afford.

Licensing

Model how the price behaves at ten times the usage.

Observability

End to end tracing, because agents keep changing.

Source Gartner, Market Guide for AI Gateways, 13 October 2025, ID G00839683

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PART 02

The Landscape







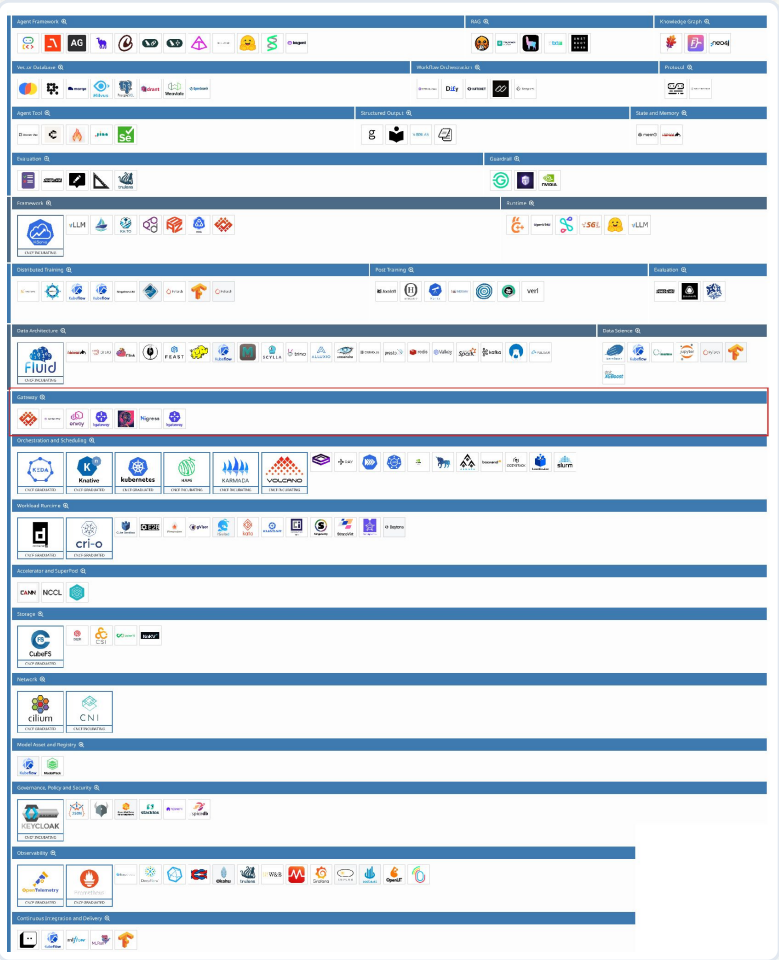




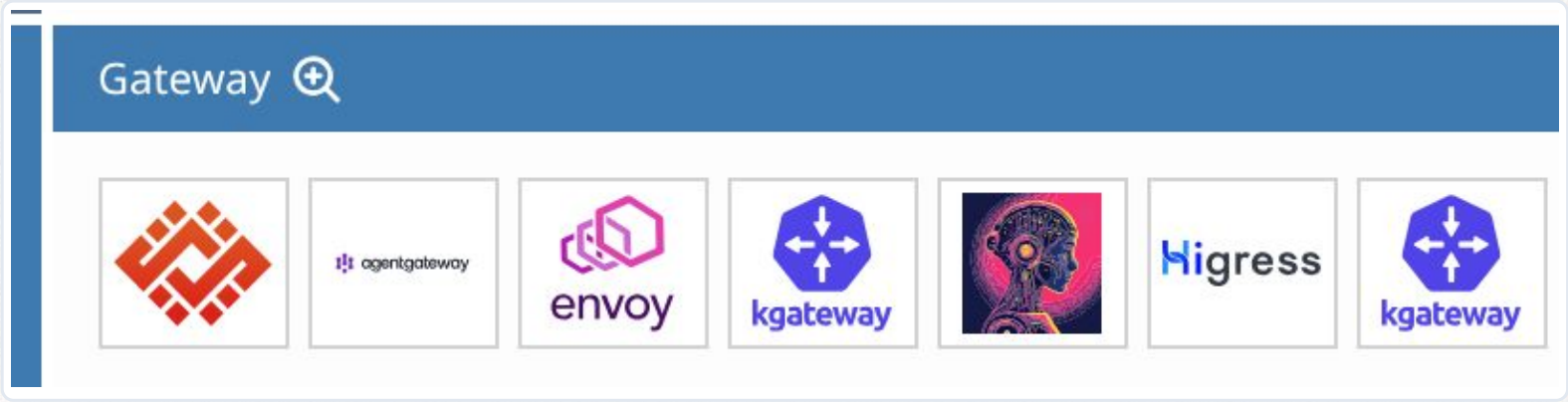




AI Native Infra and Gateway Landscape



The whole Gateway category, zoomed



6
projects in the category

77
projects in AI Native Infra

2
are CNCF projects in their own right

Six projects out of seventy seven. Compare that with Orchestration and Scheduling at seventeen. This category is young.

The six projects in AI Native Infra / Gateway

Every number re-verified against GitHub on 23 August 2026. All six are Apache 2.0

	Project	Behind it	Governance	Stars	Started	What it is
	Envoy AI Gateway	Envoy, Tetrate	Envoy subproject	1.9k	Oct 2024	Unified access to generative AI services on Envoy Gateway
agentgateway	**Agentgateway**	Solo.io, LF	Linux Foundation	4.5k	Mar 2025	One dataplane for LLM, MCP and A2A traffic
Higress	**Higress**	Alibaba	CNCF Sandbox 2026	9.2k	Oct 2022	AI native gateway on Istio and Envoy, Wasm plugins, console
kgateway	**Kgateway**	Solo.io	CNCF Sandbox 2025	5.7k	2018 as Gloo	Mature Envoy Gateway API control plane, now API focused
	Volcano Kthena	Huawei, Volcano	Volcano subproject	0.4k	May 2025	Kubernetes native LLM serving with an inference router
	MCP Gateway Registry	Agentic Community	Community project	0.9k	May 2025	Governed registry and gateway for MCP servers

Only Higress and Kgateway are CNCF projects on their own. The other four sit under a parent project, a different foundation, or no foundation at all.

Source CNCF Cloud Native AI Landscape, l.cncf.io/?group=ai-native, and GitHub, read 23 August 2026

The mental model: four layers, not one box

Most confusing arguments about AI gateways are two people talking about different layers

Client layer — Applications, assistants, agents

AI gateway — Governance. Identity, credentials, cost, guardrails

Inference gateway — Performance. Which replica, which GPU, which cache

Serving engine — Execution. Runs the model on the accelerator

vLLM is not an AI gateway and never claimed to be. Comparing vLLM to LiteLLM is comparing an engine to a traffic controller.

MCP and agent gateways sit beside the AI gateway layer, not below it.

Envoy AI Gateway and Agentgateway

The two projects built specifically for AI traffic, from the same Envoy world but with different bets

Envoy AI Gateway

Envoy Gateway plus AI. Kubernetes and Gateway API native, 16 providers.

STRENGTHS

- Real Envoy dataplane, so performance is a known quantity
- Credential injection for AWS, GCP and Azure, token aware rate limiting
- MCP support with server multiplexing, tool routing and OAuth
- Plugs into InferencePool, so one product covers both tiers

WATCH OUT

- Smallest community here at 1.9k stars, and the youngest codebase
- Kubernetes and CRDs only, no standalone mode
- Guardrails depend on external services
- No console. Everything is YAML

Agentgateway

All three Gartner targets in one Rust dataplane. Standalone or Kubernetes.

STRENGTHS

- LLM, MCP and A2A in one product, where the market is consolidating
- MCP tool federation, OpenAPI to MCP, all common transports
- Guardrails via Bedrock, Model Armor and OpenAI, CEL based RBAC
- Built in UI, and routing aware of GPU load, KV cache and LoRA

WATCH OUT

- Born March 2025. Very little production mileage
- Linux Foundation, not CNCF, so outside the CNCF maturity ladder
- Control plane moved out of kgateway in 2.3, older docs are wrong
- Small Rust extension ecosystem if you need custom logic

Source CNCF Cloud Native AI Landscape, l.cncf.io/?group=ai-native, and GitHub, read 22 August 2026

Kgateway and Higress

Two mature API gateways that arrived at AI from opposite directions, and made opposite decisions

Kgateway

Formerly Gloo, production since 2019. CNCF Sandbox since March 2025.

STRENGTHS

- By far the most battle tested code here, billions of API calls
- Full Gateway API conformance, route delegation for multi tenancy
- 275 contributors and real release discipline
- If you already run it, the AI path is incremental

WATCH OUT

- From 2.3 it handed AI and agentic features to agentgateway
- On its own it is an API gateway, not an AI gateway. You must pair it
- That split is recent, so docs and mental models lag
- Envoy and CRD complexity. A platform team product

Higress

AI native from the start. Istio and Envoy, Wasm plugins, CNCF Sandbox 2026.

STRENGTHS

- Most complete single product: LLM proxy, MCP hosting, OpenAPI to MCP
- Dozens of ready plugins and a console, useful on day one
- Wasm extensions in Go, Rust or JavaScript, not C++ filters
- Alibaba scale production origin, largest community in the category

WATCH OUT

- Center of gravity is China. Some docs and the plugin hub are Chinese first
- Images ship from Alibaba registries, most enterprises will mirror
- Istio dependency adds weight if you have no service mesh
- Plugin quality varies, and debugging Wasm in production is hard

Source CNCF Cloud Native AI Landscape, l.cncf.io/?group=ai-native, and GitHub, read 23 August 2026

Volcano Kthena and MCP Gateway Registry

Volcano Kthena

Kubernetes native LLM serving. Controller and router, deployable separately.

STRENGTHS

- Router understands inference: KV cache aware routing, PD disaggregation
- Gang scheduling and topology aware placement through Volcano
- Cost driven autoscaling with budget constraints
- Modular. Take only the router, keep your existing gateway

WATCH OUT

- Not an enterprise AI gateway. No SaaS routing, no credential brokering
- Only relevant if you self host models on Kubernetes
- Youngest and smallest in the category at 431 stars
- Full value needs Volcano, another platform component to run

MCP Gateway Registry

One governed entry point and inventory for MCP servers, agents and skills.

STRENGTHS

- Solves what nobody else does: an inventory and access model for MCP
- Per user egress OAuth, so SaaS MCP servers can be adopted centrally
- Works with Keycloak, Entra ID, Okta, Auth0, Cognito and PingFederate
- Semantic search over tools, so agents get relevant tools

WATCH OUT

- nginx, FastAPI and MongoDB. Not an Envoy shaped dataplane
- Community run with no vendor or foundation, 58 contributors
- No LLM routing and no cost control. Not your model gateway
- Documentation leans heavily on AWS deployment paths

Source CNCF Cloud Native AI Landscape, l.cncf.io/?group=ai-native, and GitHub, read 23 August 2026

The routing layer next door

Three projects people keep calling AI gateways. They are inference routers, and the difference matters

Gateway API Inference Extension

746 stars · Kubernetes SIG

Turns any ext-proc capable Gateway API proxy into an inference gateway. Defines InferencePool and the Endpoint Picker that routes to the best replica.

A standard, not a product. Endpoint Picker code just moved to the llm-d repos, so the APIs are still settling.

llm-d

4.1k stars · CNCF Sandbox

Distributed inference on vLLM and the Inference Extension. Prefix cache aware routing, prefill and decode disaggregation, KV offloading across accelerators.

A serving stack, not a governance layer. You still need something in front for identity, cost and guardrails.

vLLM Semantic Router

5.2k stars · vLLM project

A programmable mixture of models router. Picks or composes a model path per request from signals, user preference and policy, with semantic caching.

A routing brain, not a gateway. No credential brokering, no RBAC, no audit. Put it behind or inside one.

Rule of thumb: if it needs your provider API keys it is an AI gateway. If it needs your GPU metrics it is an inference router.

Source CNCF Cloud Native AI Landscape, l.cncf.io/?group=ai-native, and GitHub, read 23 August 2026

The tools people actually run, that are not in the category

Popularity and governance maturity are different axes. Know which one you are optimizing

LiteLLM

57.0k stars · since Jul 2023

STRENGTHS

- 100+ providers behind one OpenAI shaped API
- Virtual keys, spend tracking, budgets, admin UI
- MCP tools and A2A agent invocation shipped
- Fastest path to a governed endpoint. Netflix uses it

WATCH OUT

- Licence not clean open source. Enterprise is commercial
- Not Kubernetes native. You own high availability
- Config sprawls when teams share one instance

Portkey Gateway

12.8k stars · MIT · since Aug 2023

STRENGTHS

- Very fast, edge deployable, 1,600+ models
- Conditional routing, retries, fallbacks, caching
- Guardrails built into the open source gateway
- One small binary in front of SaaS models

WATCH OUT

- The OSS gateway funnels toward a commercial control plane
- Observability and governance UI are the paid product
- Weaker Kubernetes story than the Envoy options

9Router

26.1k stars · MIT · since Jan 2026

STRENGTHS

- One OpenAI compatible endpoint over 40+ providers
- Translates OpenAI, Claude and Gemini formats
- Automatic fallback on 429, plus token compression
- Aimed at coding agents: Claude Code, Cursor, Cline

WATCH OUT

- Built to save one developer money, not to govern
- No team RBAC, no audit trail, no quotas, no guardrails
- Chaining free tiers raises terms of service questions

Source CNCF Cloud Native AI Landscape, l.cncf.io/?group=ai-native, and GitHub, read 22 August 2026

The Others from Gartner

API management platforms adding AI extensions

Also Gravitee, Tyk, F5, Sensedia, Lunar.dev, API7, Workato, Cequence



Best when you already standardize on API management. The AI features are newer than the product, so test them rather than assume them.

AI platforms with a built in gateway

Also TrueFoundry, nexos.ai, Airia, SS&C Blue Prism



Best when your teams already live on that platform. The gateway is a feature of the platform, so leaving one means leaving the other.

Purpose built open source or open core

Also APISIX AI Gateway, Traefik AI Gateway, Helicone, NeuralTrust, Lasso



Best when you want a neutral, provider agnostic gateway with good developer ergonomics. This is where the CNCF projects sit.

Source Gartner, Market Guide for AI Gateways, 13 October 2025, ID G00839683

Comparison

Read from public documentation in August 2026

Project	Access control	Token cost control	Multi provider	Observ ability	Guard rails	MCP	A2A	Self hosted routing
Envoy AI Gateway	●	●	●	●	◐	●	○	●
Agentgateway	●	●	●	●	●	●	●	●
Higress	●	●	●	●	●	●	○	◐
Kgateway	●	○	○	●	○	○	○	●
Volcano Kthena	○	◐	○	●	○	○	○	●
MCP Gateway Registry	●	○	○	●	◐	●	◐	○
LiteLLM	●	●	●	●	●	●	◐	◐
Portkey Gateway	●	●	●	●	●	◐	○	○
9Router	○	◐	●	◐	○	○	○	○

● built in ◐ partial, or via plugin or external service ○ not in scope for the project

A row of green dots is not a recommendation. Agentgateway scores highest and has the least production mileage.

PART 03

The Choice



Subscriptions or a gateway?

Buy vendor subscriptions

Claude Team, ChatGPT Business, Gemini for Workspace

What you get	A finished application. Chat interface, IDE plugins, mobile. Priced per seat per month.
Who it serves	People typing. Humans in a browser or an editor.
Governance	The vendor's own admin console. SSO, retention settings, an audit trail of that vendor's app.
Cost shape	Fixed per seat. Predictable, and capped by the vendor's rate limits rather than by you.
Where it stops	It governs that vendor's product. It cannot see one line of what your own code sends to an API.

Call the API through a gateway

Your applications and agents

What you get	A control point. You still build the application.
Who it serves	Software. Services, agents, batch jobs, anything programmatic.
Governance	One point across every provider. Per team budgets, key custody, guardrails, one audit trail.
Cost shape	Per token. Variable and able to spike, which is exactly why the budget controls matter.
Where it stops	It does nothing for a person using a chat website. That traffic never passes through it.

Most organisations need both. The subscription governs people, the gateway governs software, and neither substitutes for the other.

When not to use an AI gateway

One app, one provider

Low volume, a single team, no plan to switch. The provider SDK is simpler and the gateway earns nothing.

Your AI use is people, not code

If everyone is in a chat window, buy seats and configure the vendor console. A gateway would sit on a path nobody uses.

You already run API management

Kong, Apigee, APIM and the rest are adding AI features. Gartner's explicit advice is to exhaust what you own first.

Latency is the product

An extra hop you gain nothing from. If there is no routing, no budget and no guardrail to apply, do not add one.

Nobody owns it

An unowned gateway becomes a single point of failure between every application and every model. That is worse than no gateway.

You are still prototyping

Do not build a control plane before you have traffic worth controlling. Add it when the second team appears.

Which gateway for which job

Which boundary are you governing?

You call SaaS models only

No GPUs. You need identity, keys, budgets and an audit trail across providers.

Start with LiteLLM or Portkey. Move to Envoy AI Gateway or Higress when you need Kubernetes native policy.

You self host models

GPUs in your own cluster. Replica choice, KV cache reuse, keeping accelerators busy.

Inference Extension with vLLM, or Kthena, or llm-d. Then put an AI gateway in front for governance.

You are building agents

MCP servers multiplying, each with its own credentials and its own tool policy.

Agentgateway for one dataplane over models and tools. MCP Gateway Registry when the problem is inventory.

You already run an API gateway

Kong, APISIX, Apigee or kgateway is in production and the platform team owns it.

Test its AI extensions first. Gartner says exhaust what you own before buying a second control plane.

A reference shape that survives most requirement changes

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Clients

Apps, assistants, agents

Tier one gateway

Identity, top level routing, quotas, cost attribution, guardrails



agentgateway
Higress

Tier two gateway

Endpoint picking, KV cache and LoRA awareness





Model serving

Runs the model on your accelerators

vLLM
SGL


Beside tier one: MCP and agent gateway

Same identity, same policy engine, same audit trail, applied to agent to tool and agent to agent traffic.

SaaS providers



aws

reached with injected credentials

Live Demo



Wrap Up



Key takeaways

01

Classify by target, not by brand

Model, MCP or agent gateway. Name the boundary and the shortlist writes itself.

02

The floor is governance

Access, cost, routing, observability, security. Missing one means it is not a gateway.

03

The category is young

Six projects, two of them CNCF in their own right, none of them a default.

04

Keep the layers apart

Governance at tier one, performance at tier two, execution in the engine.

05

Prove it before you standardize

Run the proof of concept and measure the added latency. You will upgrade often.

An AI gateway is where your organization decides what its AI is allowed to do, and how much it is allowed to spend doing it.

Sources Gartner Market Guide G00839683, Gartner GTP webinar 856252, CNCF Cloud Native AI Landscape and GitHub, read 22 August 2026

Need Contributors: Kubernetes Docs in Thai

Opsta

A community proposal from K.Benz Jaruwat Panturat

There is no Thai version of the Kubernetes documentation today. The idea came out of a KubeCon Japan session on docs localisation, and the proposal is to start one. We need two more people.

People learn faster in their own language

The docs are the front door to Kubernetes. Reading them in Thai removes a step for everyone not reading English comfortably.

Docs are the easiest way into open source

A translation PR is a real contribution with a low barrier, and a good way to put engineers through the OSS process for the first time.

Translating is how you find the gaps

You cannot translate a page you do not understand. People come out knowing Kubernetes better than when they went in.

It needs at least two contributors before it can go to the SIG. Nobody can approve their own pull requests.

Interested? Join the channel

That is the only action. No sign up sheet.

slack.cncf.io

the official CNCF Slack invitation page

1. Tick both boxes, then Join Slack
2. You land in the cloud-native workspace
3. Find the #thailand channel, join, say hello



MARCH 2027

KCD

GROWING CLOUD NATIVE TOGETHER